
# CHAPTER

# ATA-54

# NACELLES

**THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY**

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# Contents

## Nacelles - General

### Introduction

The nacelle makes an aerodynamic surface around the power plant. The power plant, exhaust assembly, wing structure, and main landing gear are also contained in the nacelle.

The forward nacelle extends forward from the wing front spar frame and contains the main support structure for the propulsion system.

The nacelle center section extends from the nacelle forward section to the lower aft section.

The nacelle aft section extends aft from the aft spar.

### General Description

*Refer Figure 54-1*

Each nacelle has the three parts that follow:

- Forward Nacelle
- Center Nacelle
- Aft Nacelle

The forward nacelle encloses the engine mounting structure that supports the propeller, gearbox, engine, systems and cowlings.

Engine operating and aerodynamic loads are transferred through the structure to the nacelle center section and wing box.

The nacelle center section houses the main landing gear and wing nacelle attachments points.

The nacelle aft section houses part of the exhaust assembly, and the main landing gear wheels when it is retracted. It includes the structure of the lower fairing and the forward and aft upper fairings.

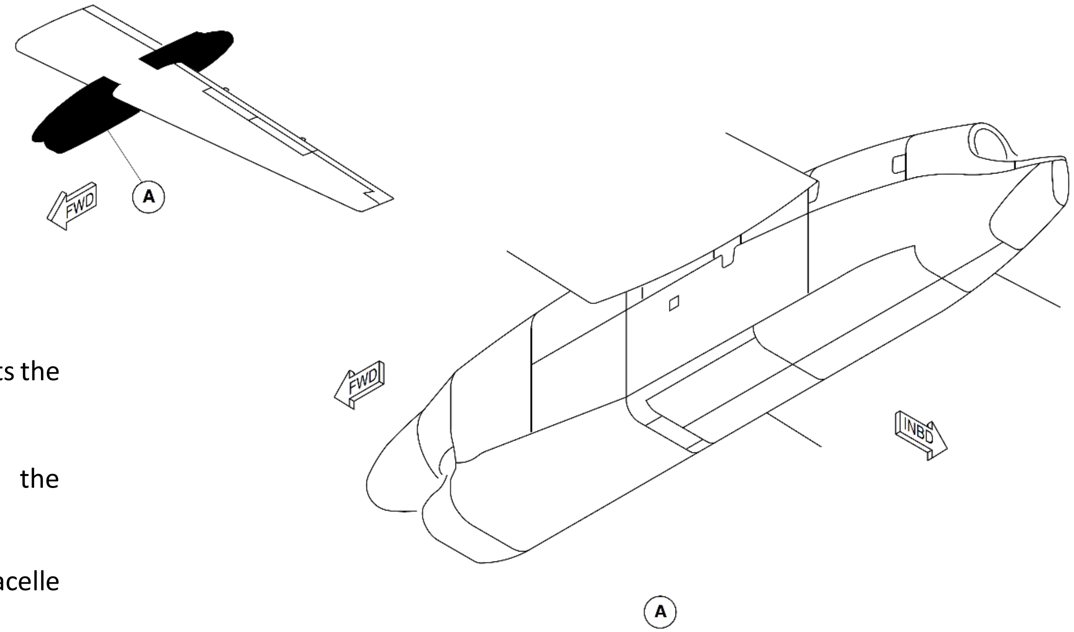



**Figure 54-1 Nacelle - Location**

**Detailed Description**

Refer Figure 54-2

Each engine is housed in a nacelle installed below the wing.

The nacelle houses the components that follow:

- Main landing gear
- Engine exhaust
- Landing gear accessories

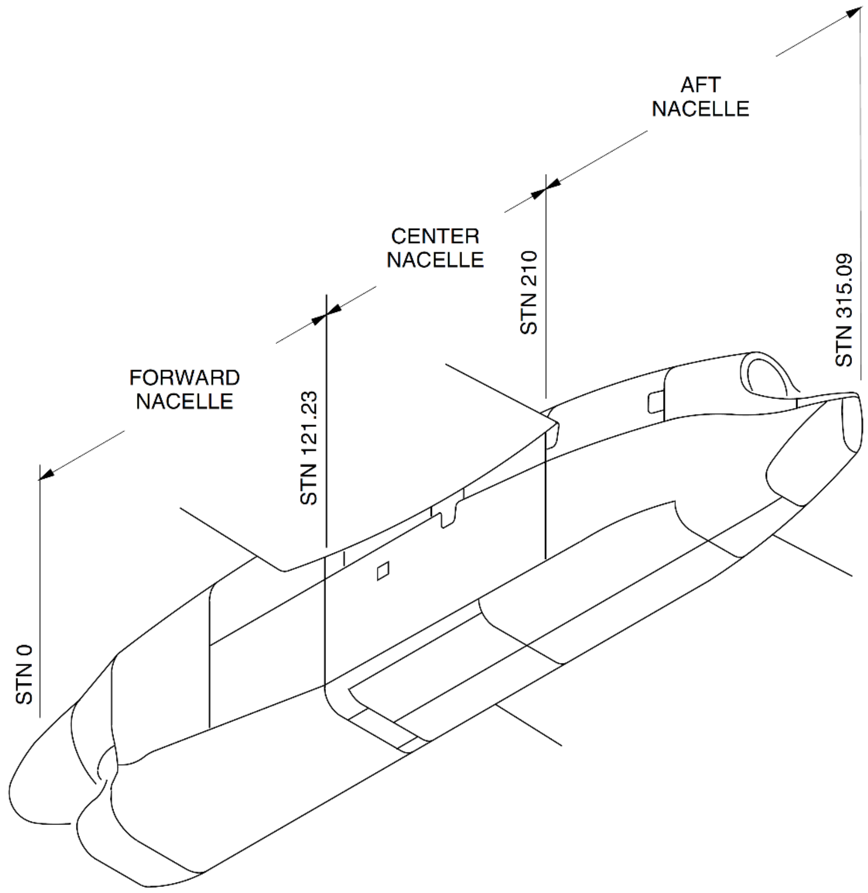



**Figure 54-2 Nacelle**

## Forward Nacelle

Refer Figure 54-3

The forward nacelle is a housing for the power plant. It includes the structure of the engine mount system, fairings and access doors, and lower cowl.

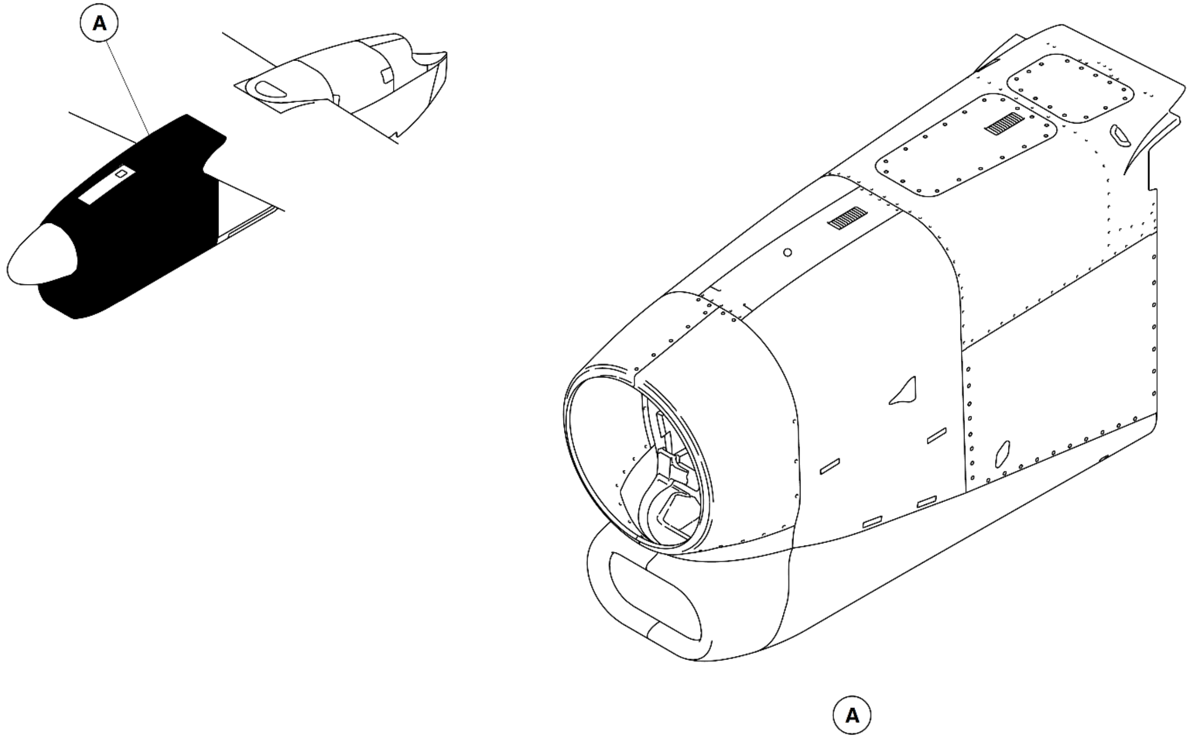



**Figure 54-3 Forward Nacelle - Location**

**Center Nacelle**

*Refer Figure 54-4*

The center nacelle is a housing for the wing structure, the forward part of the exhaust assembly, and where the main landing gear attaches. It includes the structure of the ’A’ frame, two main side frames, and rear bulkhead.

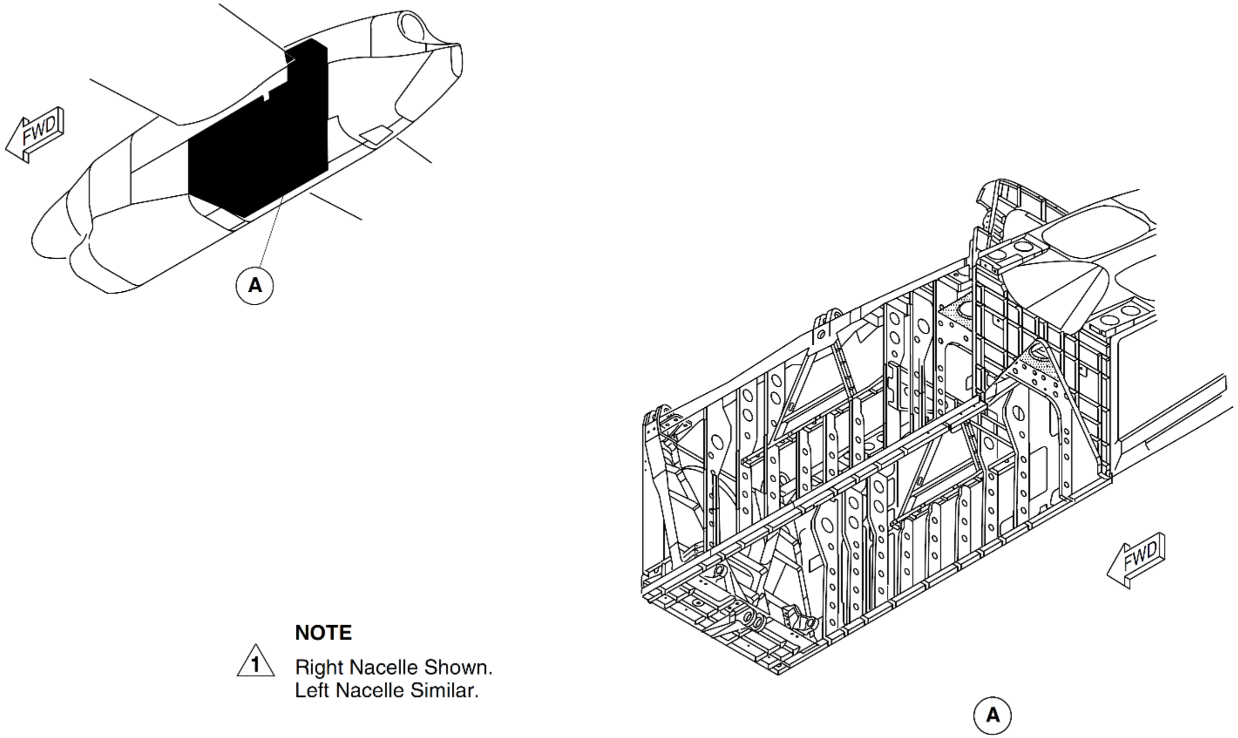



**Figure 54-4 Center Nacelle - Location**

## Aft Nacelle

*Refer Figure 54-5*

The aft nacelle is a housing for the aft part of the exhaust assembly, and the main landing gear wheels (when retracted). It includes the structure of the lower fairing and the forward and aft upper fairings.

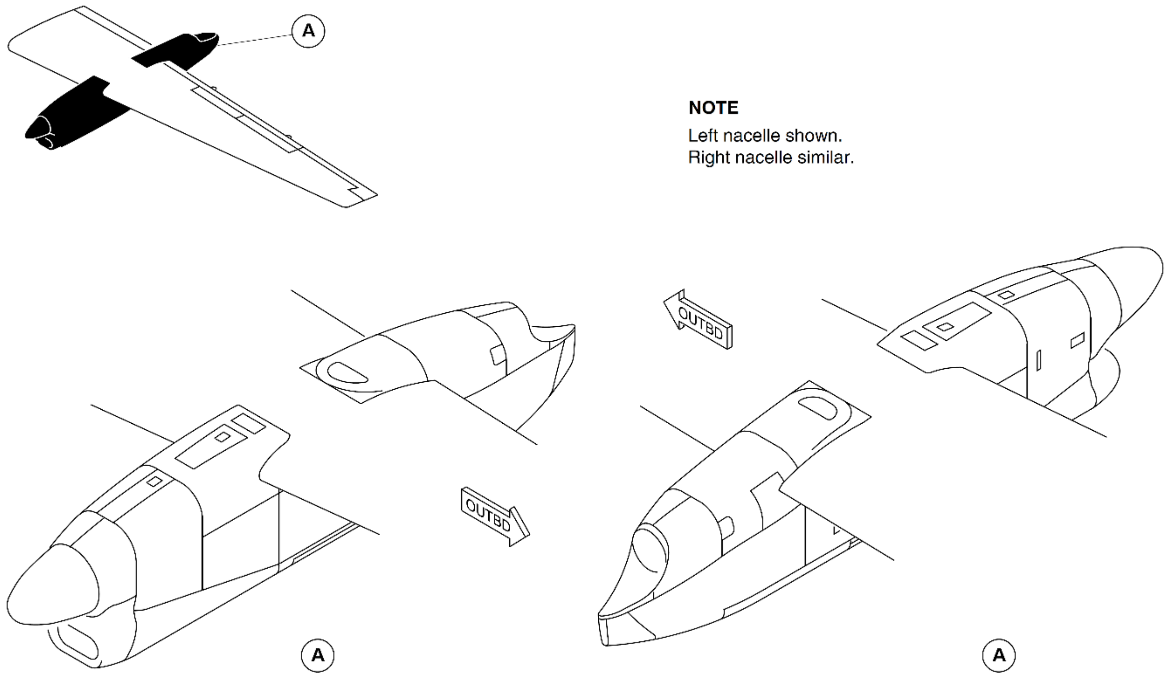



**Figure 54-5 Aft Nacelle - Location**

# Forward Nacelle

## Introduction

The nacelle encloses the engine mounting structure that supports the propeller, gearbox, engine, systems and cowlings. Engine operating and aerodynamic loads are transferred through the structure to the nacelle center section and wing box.

## General Description

Refer Figure 54-6

The forward nacelle extends forward from the wing front spar frame and contains the main support structure for the propulsion system.

The Forward Nacelle has the components that follow:

- Forward Nacelle Access Panels
- Forward Nacelle Fairings
- Forward Nacelle Struts

Engine torque is transferred to the forward frame by the Hydraulic Torque Compensation System.

The forward frame is designed to redistribute loads in the event of a mount or strut failure.

The struts distribute the load from the forward frame through the mid frame to the center section and wing box using fittings and bolted joints. The mid frame also supports the rear of the engine through the two aft elastomeric vibration isolator mounts.

The forward frame has the interfaces that follow:

- Engine vibration isolators
- Hydraulic torque compensation system
- Three strut attachment brackets
- Ground support engine hoist bracket
- Lower cowling attachment
- Eight engine bonding leads
- Upper forward cowling location pin
- Spine attachment
- Electrical loom attachments
- Side door stay attachment
- Engine and accessories clearance.

The mid frame is used to attach the rear engine mounts and distribute the loading from the engine and propeller. It is a machined titanium frame and has the interfaces that follow:

- Engine aft vibration isolators
- Pneumatic pre−cooler and two valves
- Airframe de−ice attachment
- Fuel feed and motive flow line terminations
- Hydraulic feed, return and case drain line terminations
- Fire extinguisher line terminations
- Harness clipping
- Strut fittings
- Lower cowl connection
- Leading edge fairing connection
- Spine connection
- Forward door interface and door stay latch
- Aft door attachment
- DC fault current ground lead.

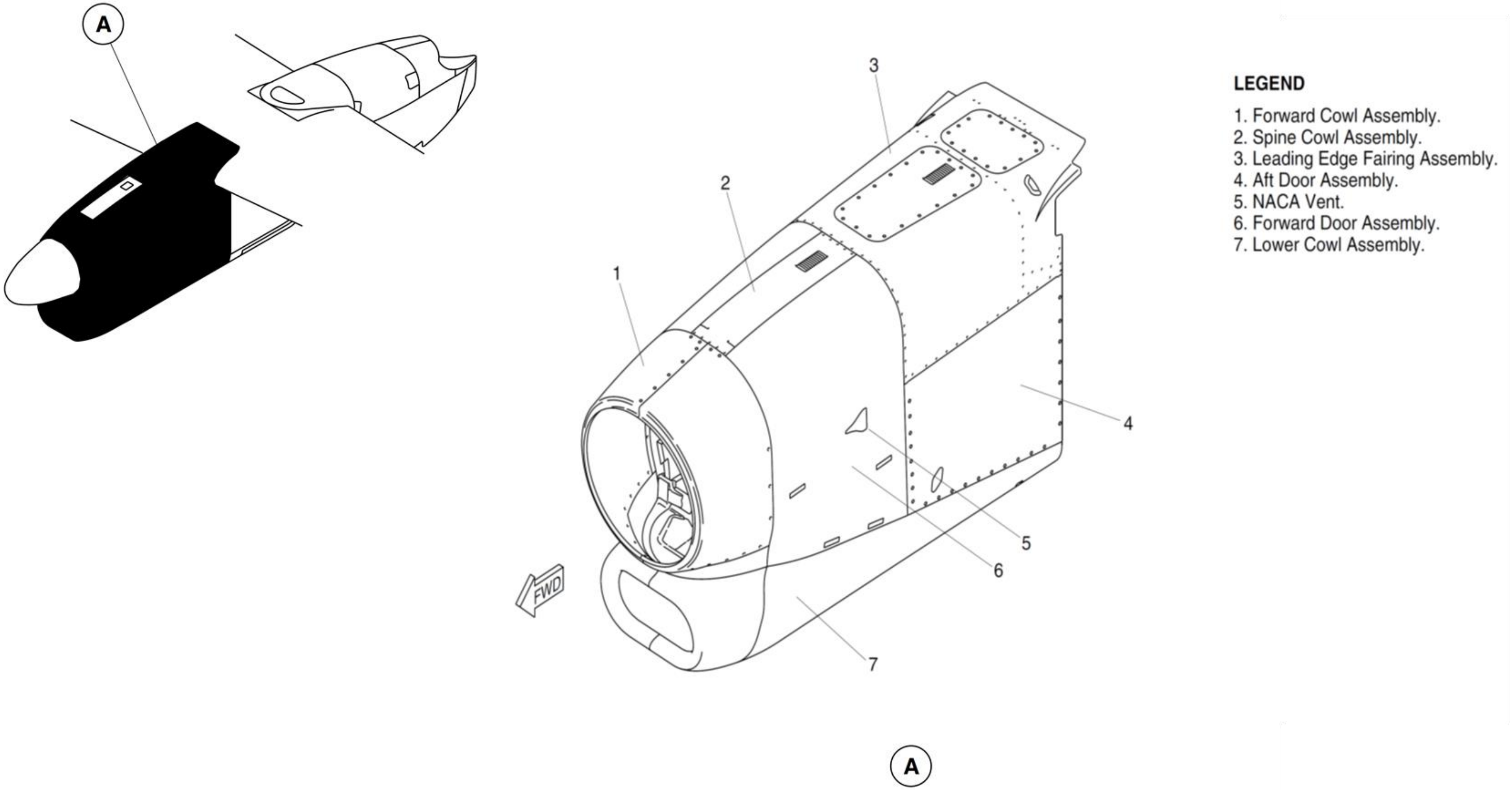



**Figure 54-6 Forward Nacelle**

**Detailed Description**

*Refer Figure 54-7*

The engine mounting structure has a forward frame (Horse Collar) which supports the engine through three elastomeric vibration isolator mounts, to transfer the loads from the propeller and engine into the struts through strut fittings. The frame is a machined titanium section with an integral Warren Girder web stiffening section that is installed perpendicular to the engine axis, and is located external of the engine rotor burst zone.

The engine isolation system has five mounts:

- Three located on the forward face of the forward frame
- Two located on the aft face of the mid frame.

The five mounts are elastomeric and help reduce the transmission of propeller vibrations to the cabin interior. The isolation system limits maximum engine deflections, and provides isolation under normal operation.

The torque restraint system has two interconnected opposite acting actuators that are attached through a linkage mechanism to the forward side mount engine attachment brackets. The system is self-contained, designed to replenish and bleed itself, and contains approximately 12 in$^3$ (197 mm$^3$) of MIL−H−5605 hydraulic fluid. The cylinders and fluid lines are manufactured from stainless steel and the reservoir is aluminum alloy. In the event of a failure of the torque reaction system, the side mounts will snub and react to torque.

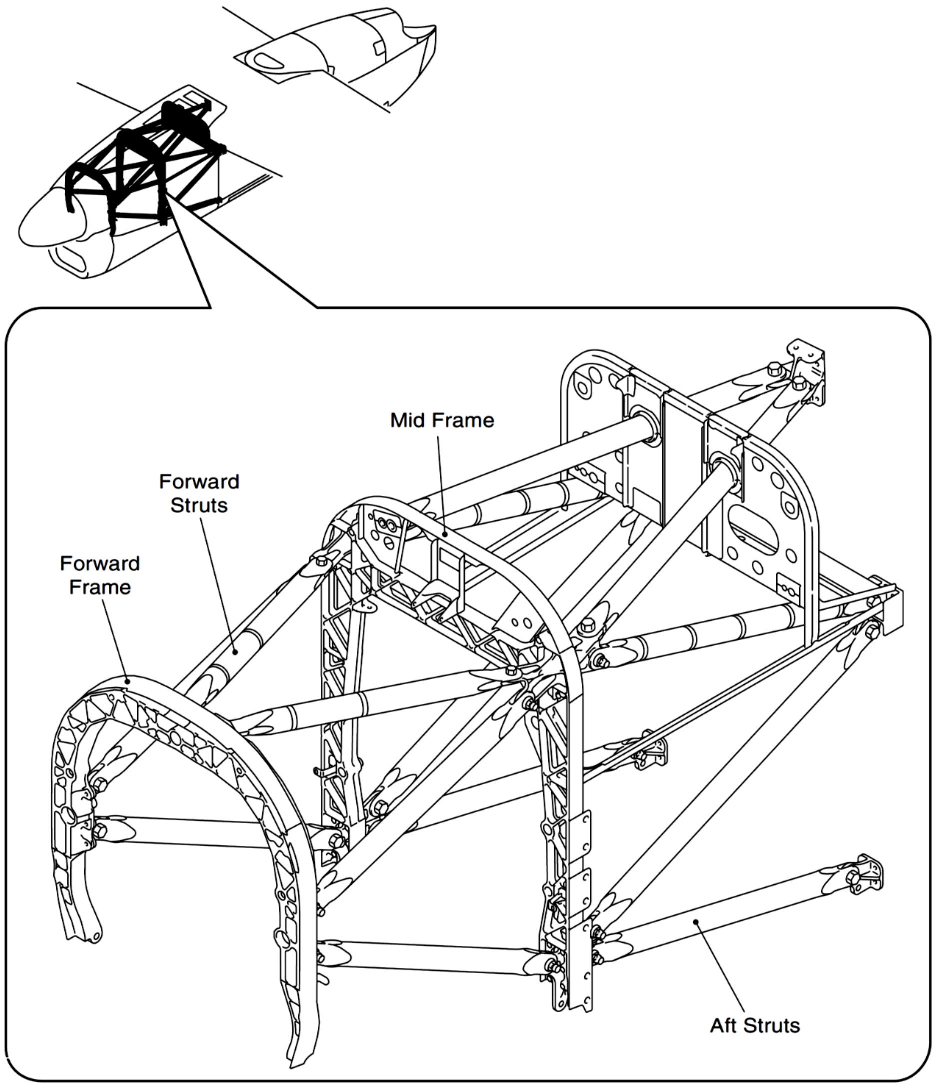



**Figure 54-7 Nacelle Support Frames and Struts**

**Forward Nacelle Access Panels**

*Refer Figure 54-8*

The upper forward cowl is manufactured from carbon epoxy composite, with copper mesh High Intensity Radio Frequency (HIRF) and Lightning protection provided on the outer surface. The cowl is attached using quick release fasteners at the bottom edge, with a location pin at the top center line position. The cowl is sealed on its aft and lower edges using a non−conductive elastomer seal bonded in place.

The upper forward cowl is removed to access the items that follow:

- Propeller back plate
- Brush blocks
- Engine mounts
- Torque reaction system
- Removal of the lower cowl.

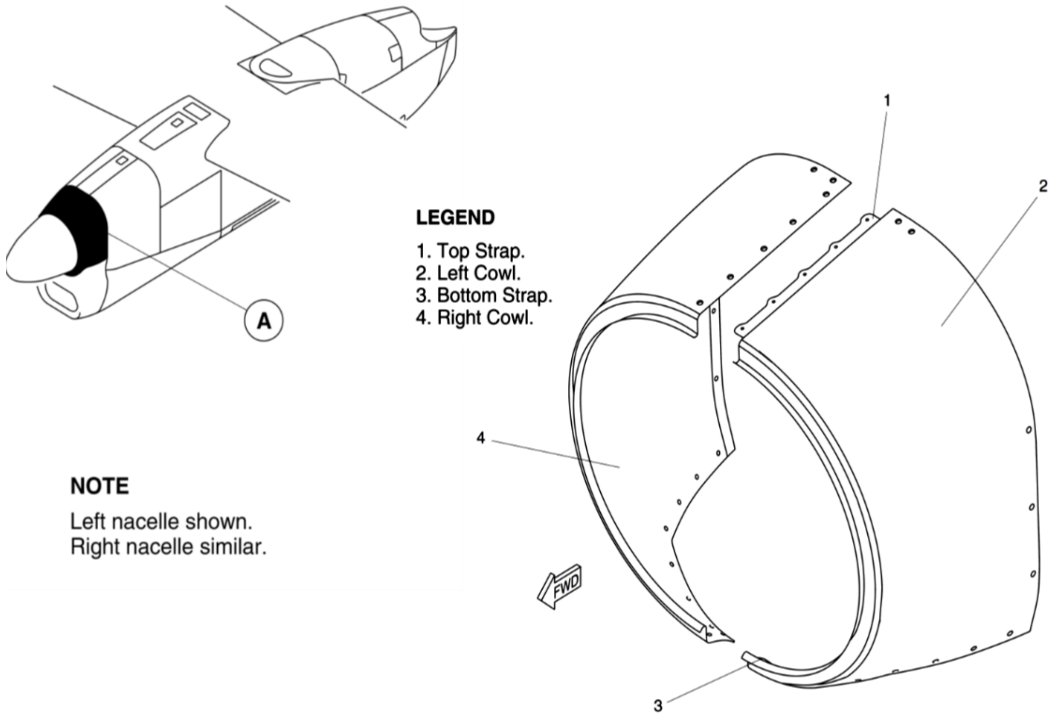



**Figure 54-8 Forward Cowl**

*Refer Figure 54-9*

The left and right forward doors are made from carbon epoxy composite with integral foam filled stiffening ribs. HIRF and lightning strike protection is provided using copper mesh on the outer surface of the doors. The doors are attached to the upper spine member, have a stay strut at each end and are fastened closed by five quick release pin latches. The left door has provision for ventilation by a NACA duct manufactured from carbon epoxy and bonded to the door inner skin. The right door has an inlet and outlet for the starter generator cooling.

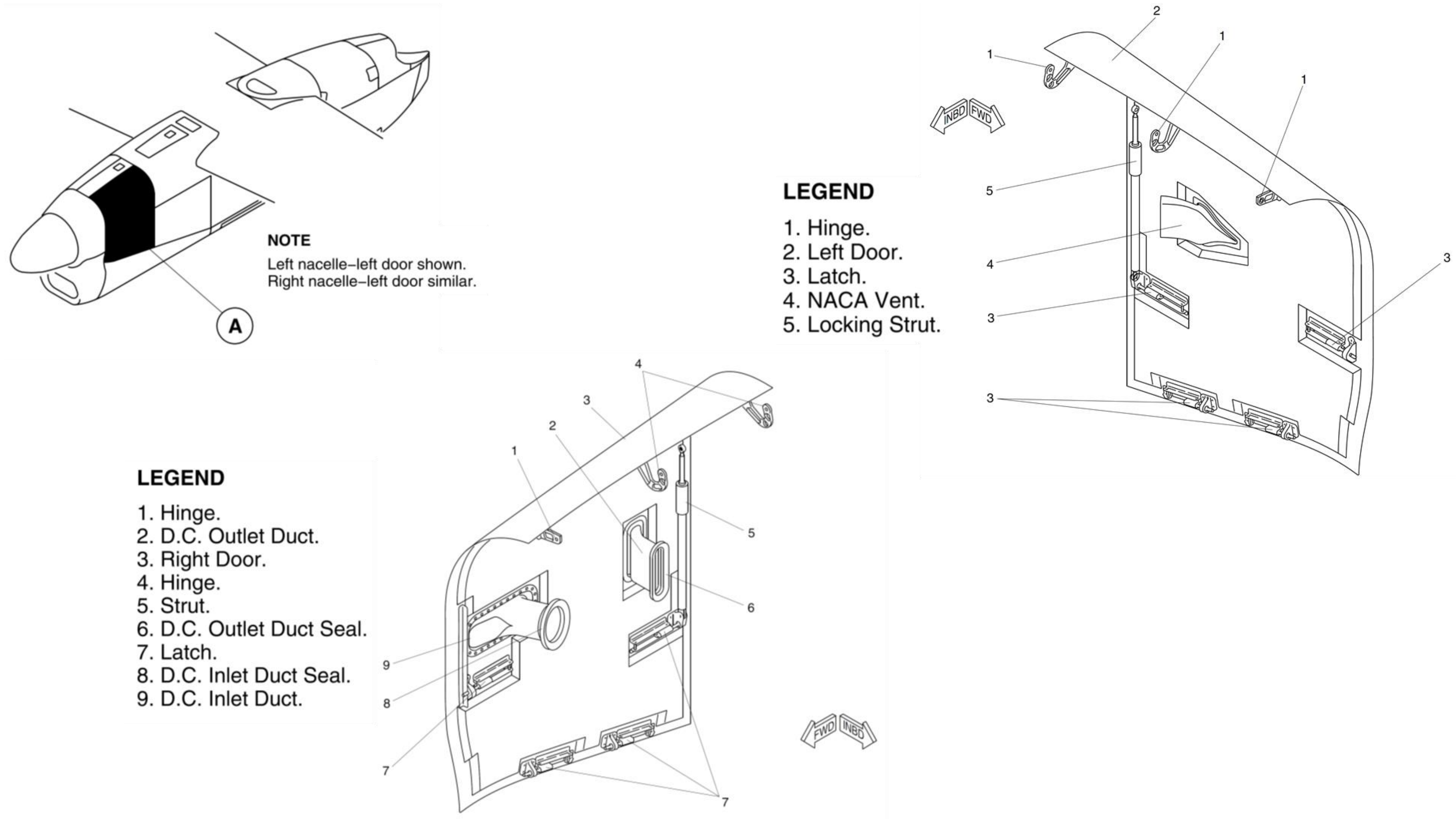



**Figure 54-9 Nacelle Forward Doors**

Refer Figure 54-10

The left and right aft doors are made from a titanium fabricated construction, with a flat outer skin and Z shaped vertical stiffeners on the inner surface. The doors are attached on the aft edge with a door open restraint on the nacelle center section. The door is fastened closed using quick release fasteners on three edges.

The left door has an outlet grille for ventilation of the upper equipment bay. A ventilation duct is installed down the firewall and ends in a P seal to give a sealed interface when the door is closed. The lower stiffener of the door has drain holes to release leaked fluid overboard to the lower cowl drain lines. The hinged aft edge of the door is sealed to prevent fluid leakage.

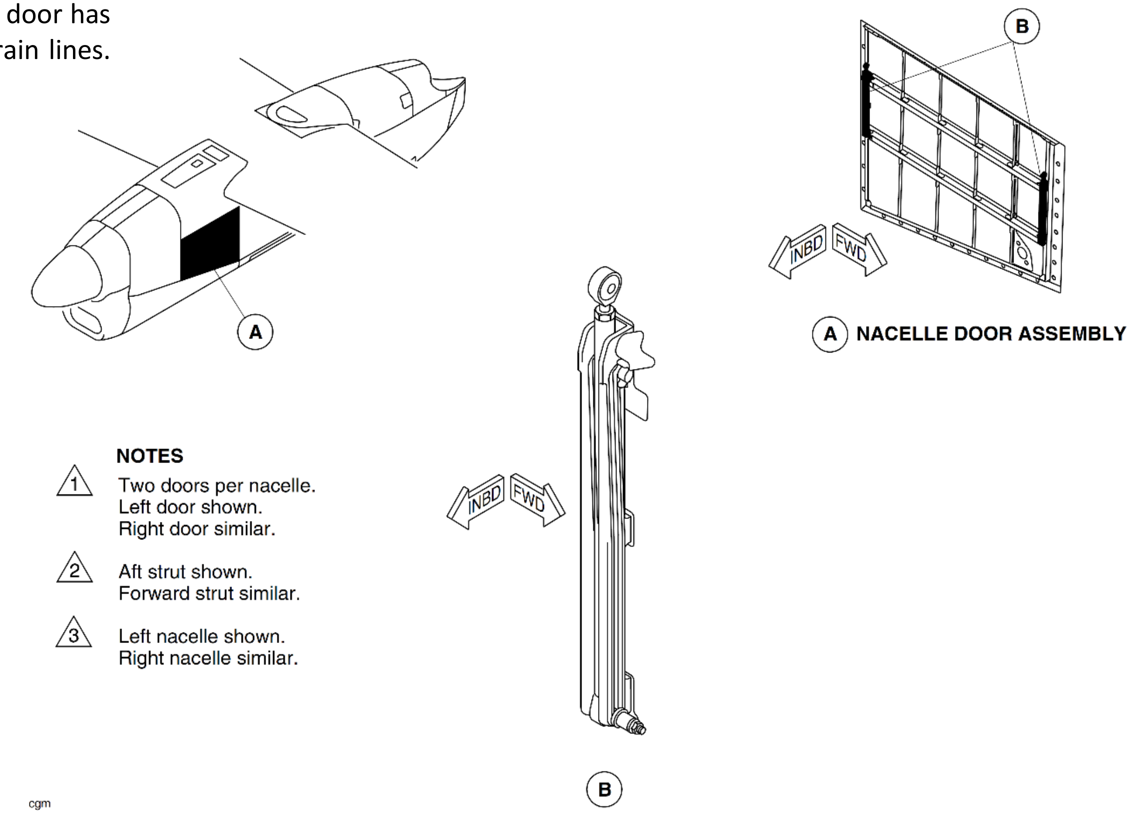
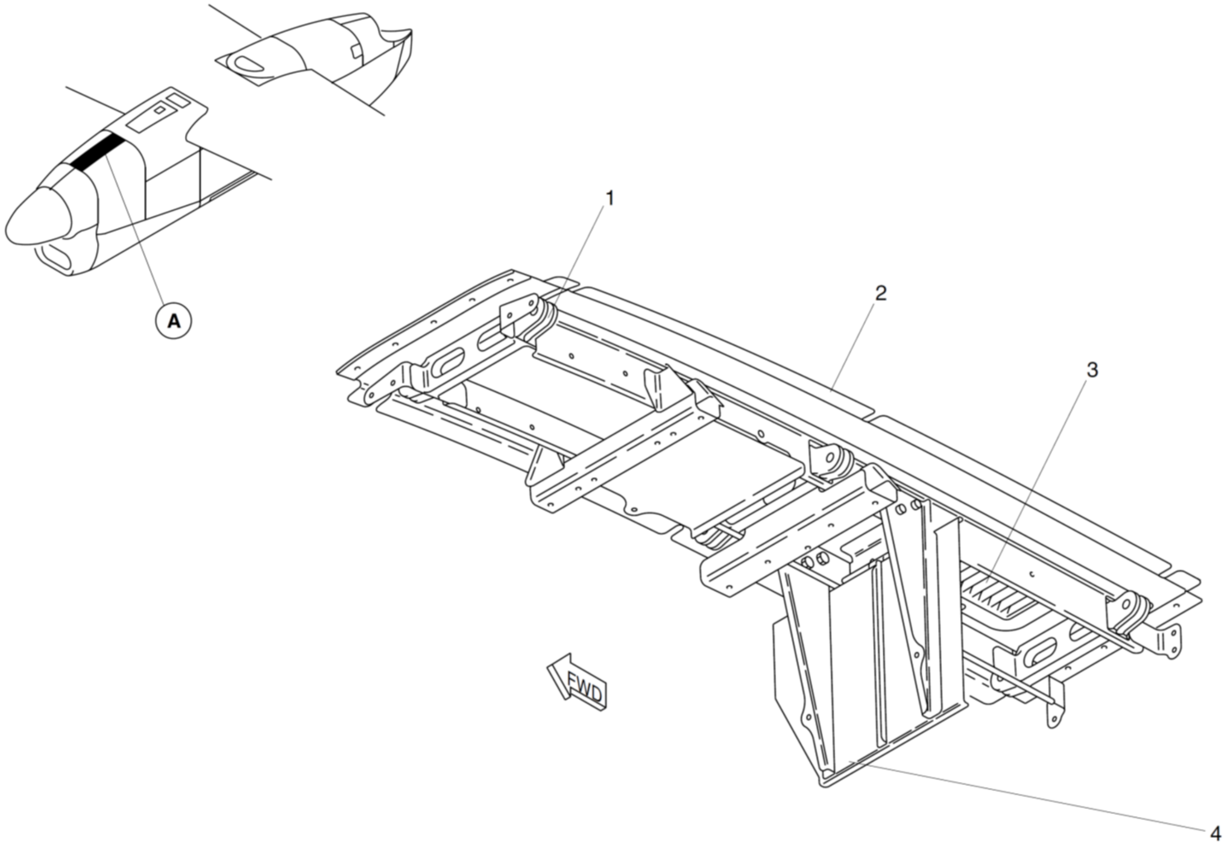



**Figure 54-10 Nacelle Side Doors**

*Refer Figure 54-11*

The spine is located on the nacelle top center line, installed between the front and mid frames, and is made from fabricated titanium with three hinge points for forward doors attachment. The spine does not carry primary structural loads, but will carry aerodynamic loading in flight. The spine acts as a hinge beam for the forward doors, and gives an attachment point and exhaust outlet for the pneumatic system precooler. The Propeller Electronic Controller (PEC) is also installed on the spine.

The leading edge fairing is made from carbon epoxy composite with foam stiffening. Copper mesh on the outer surface gives HIRF and lightning strike protection. The fairing has an access panel for access to the upper area of the zone, and leading edge bay for maintenance.




**LEGEND**

1. Typical Hinge.
2. Spine Cowl.
3. Louvre.
4. Pre−Cooler Heat Shield Assembly.

**Figure 54-11 Nacelle Spine Cowl**

Refer Figure 54-12

The lower cowl is the only primary structural cowl, and serves as a partial load path for propulsion and inertial loads during normal operating conditions. In the event of primary structural element failure, the lower cowl is designed to transfer loads to the wing box.

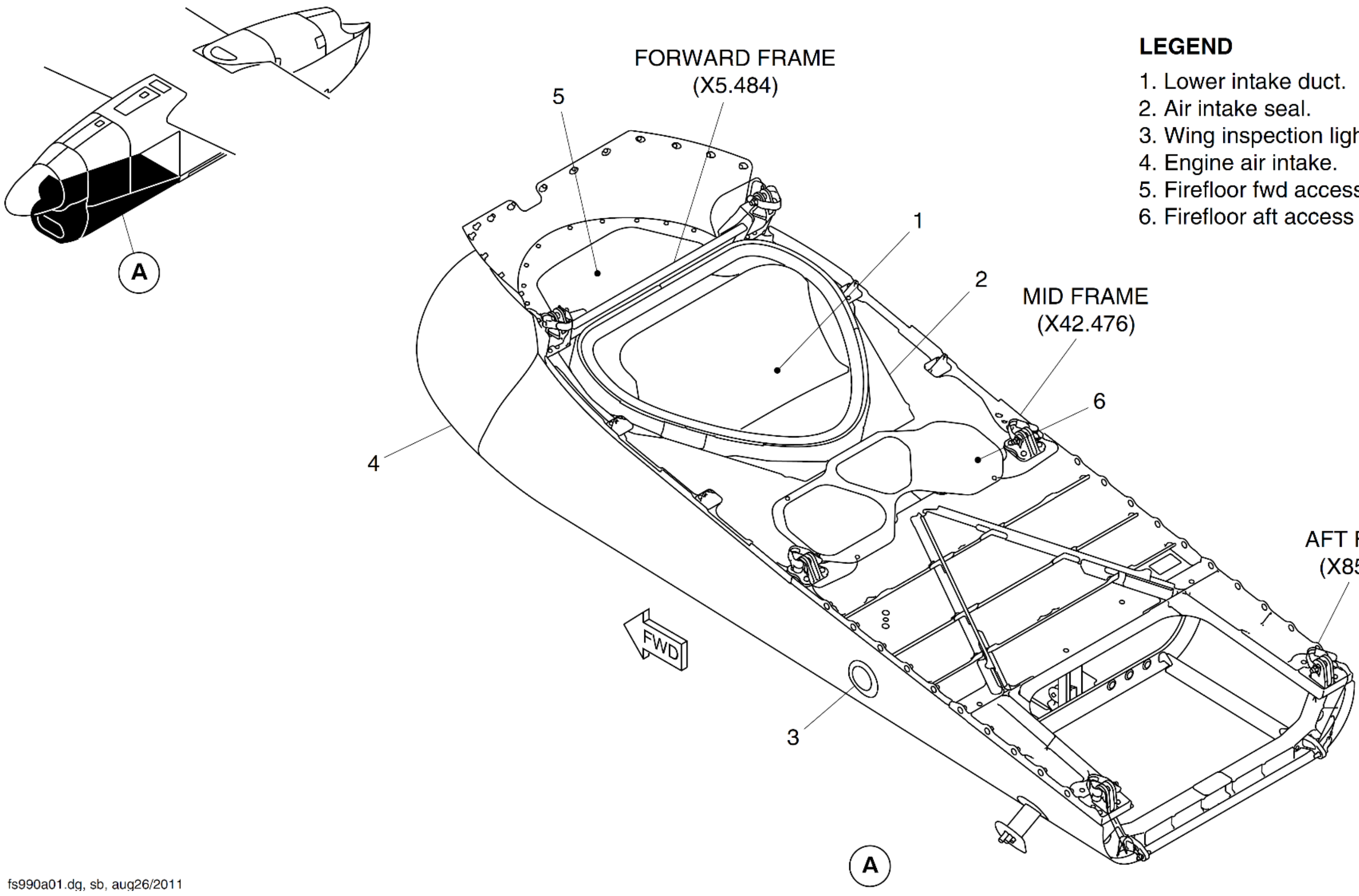
The lower cowl is an aerodynamic fairing manufactured from titanium and carbon fiber. It provides an alternative load path in case of failure within the engine mount system. It is attached to the nacelle by hinges and latches.

The lower cowl has three frames located at the major load points. These frames are contoured to match the aerodynamic shape of the cowl. The forward frame is manufactured from two titanium and two aluminum alloy sections. The mid frame is manufactured from three aluminum alloy sections, while the aft frame is a single titanium machining.

The external skin and intake duct are molded in a carbon fibre/nomex honeycomb to form the aerodynamic surface of the cowl.

The fire floor is an assembly of machined and fabricated titanium components. It attaches to the frames and skin and provides additional stiffness. The air intake is integral to the cowl and is manufactured from a fireproof monolithic laminate.

The fire floor forward access−panel is installed in the forward nacelle at the forward edge of the lower cowl. The fire floor aft access−panel is installed in the forward nacelle at the middle of the lower cowl.




**LEGEND**

1. Lower intake duct.
2. Air intake seal.
3. Wing inspection light.
4. Engine air intake.
5. Firefloor fwd access panel.
6. Firefloor aft access panel

**Figure 54-12 Nacelle Lower Cowl**

Refer Figure 54-13

Two nacelles de–icing boots (one per nacelle) form the air intake lips to remove ice buildup. The air for the nacelles de–icing boots is independently supplied.

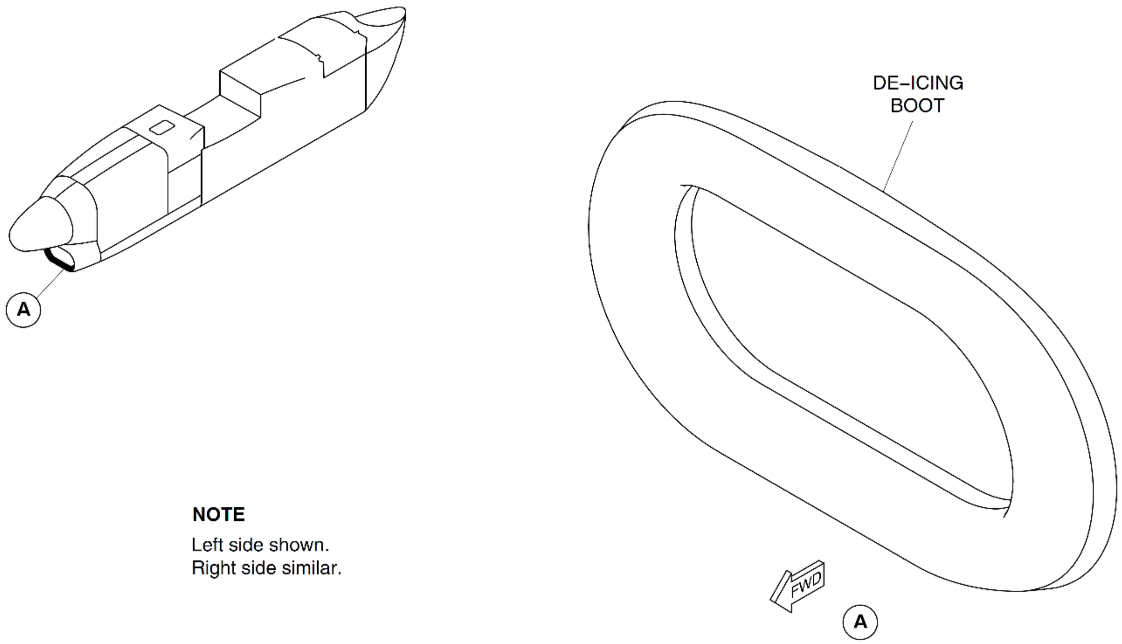



**Figure 54-13 Nacelle Deice Boots – Location**

Refer Figure 54-14 & Refer Figure 54-15

The lower cowl floor provides a natural downward flow path for fluids towards the aft end of the nacelle. At the aft end of the floor a drain sump is located spanning the width of the nacelle. Bulk drains are provided at this location to allow fast and safe dispersion of the leaked fluid through the drain mast.

The forward inlet−duct drain holes are located in the lower nacelle acoustic liner. The drain holes are in the bottom surface of the engine inlet duct (between the de−icing boot and the cowl beam).

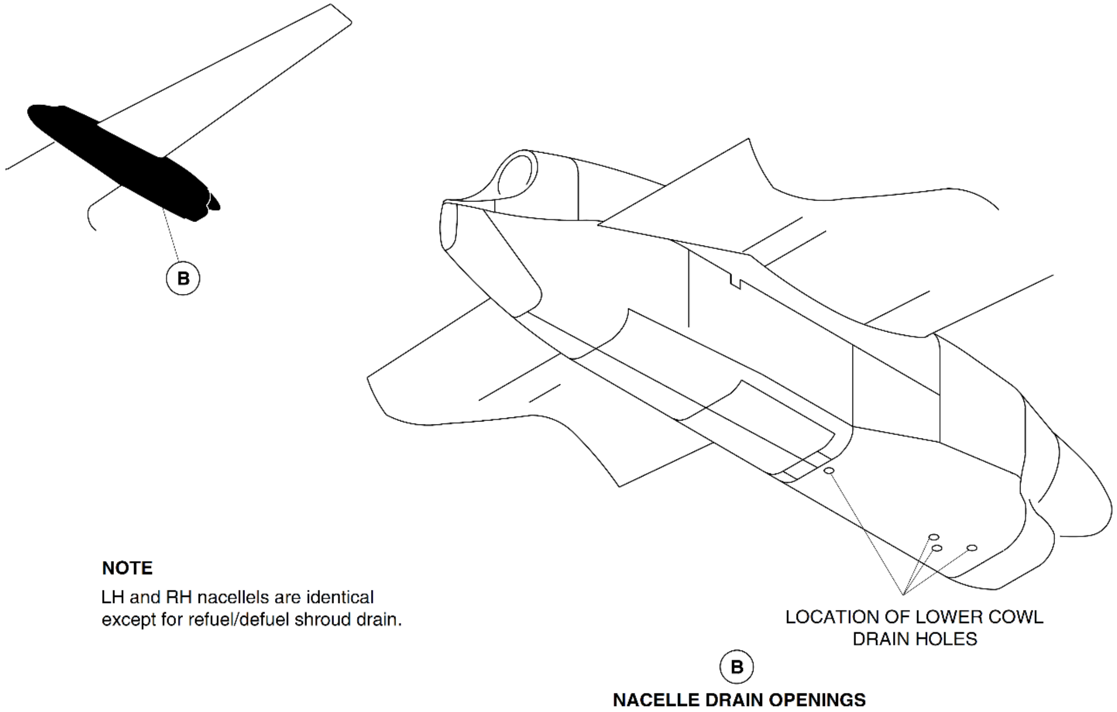



**Figure 54-14 Forward Nacelle Lower Cowl Drain - Location**

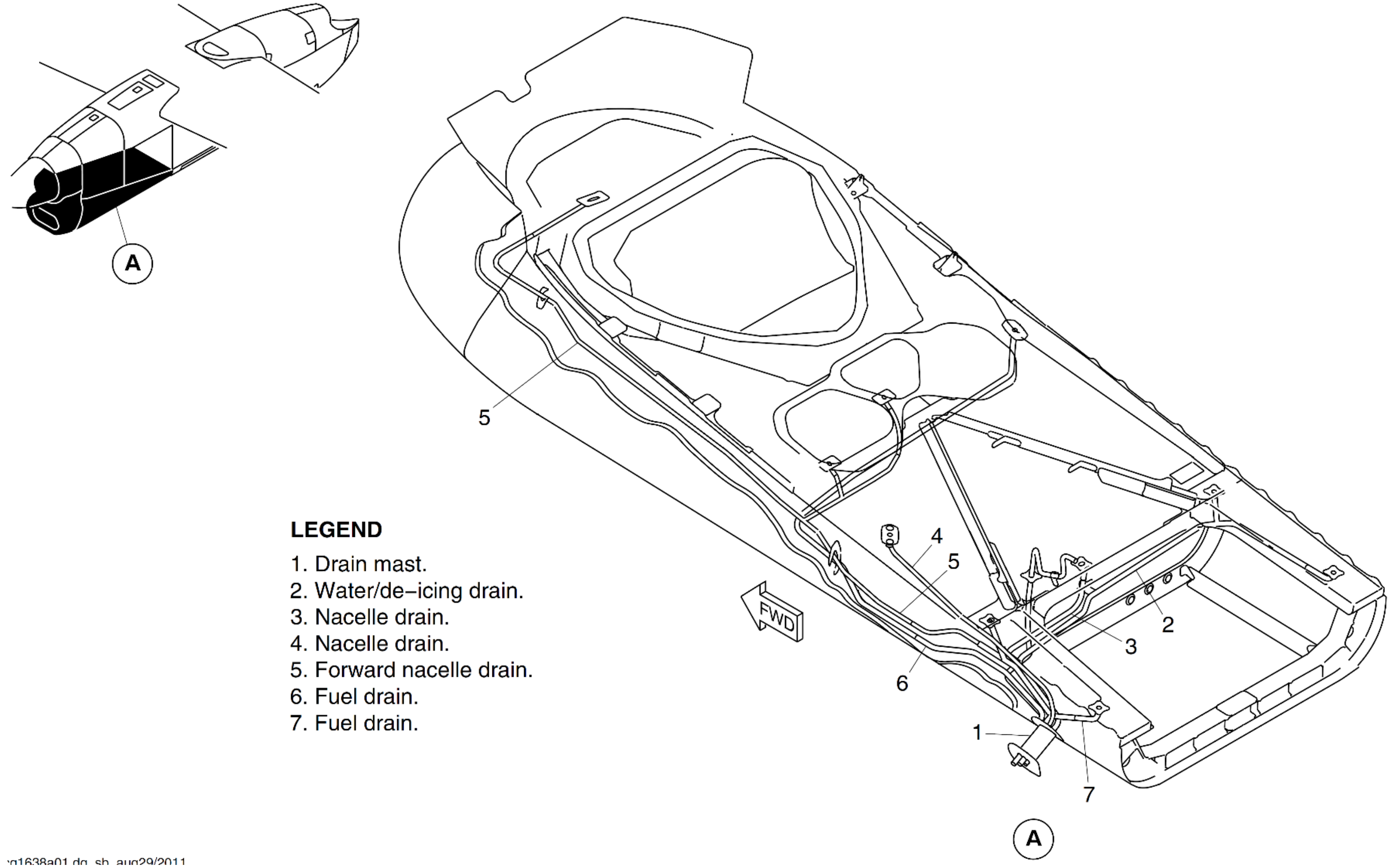



**Figure 54-15 Forward Nacelle Lower Cowl Drain**

Refer Figure 54-16

The lower cowl of each nacelle also has a Foreign Object Damage (FOD) outlet.

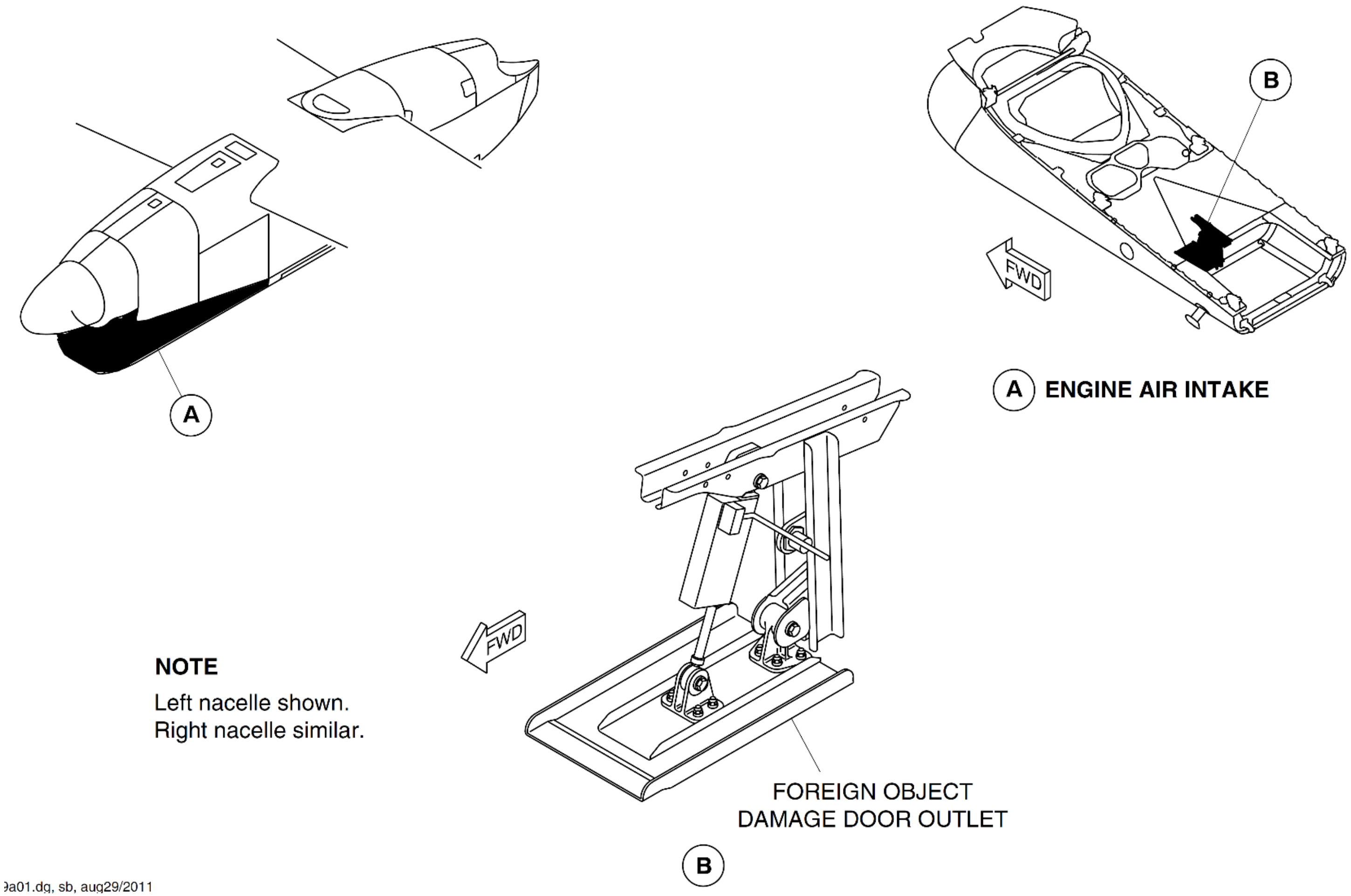



**Figure 54-16 Foreign Object Damage (FOD) Outlet**

## Center Nacelle

### Introduction

The nacelle center section houses the main landing gear and wing nacelle attachments points.

### General Description

The nacelle center section extends from the nacelle forward section to the lower aft section.

The nacelle center section, has the components that follow:

- Main Side Panels
- Main Landing Gear A Frame
- Wing Front and Rear Spar attachments
- Firewall and Oil Cooler Support Structure
- Nacelle Centre Access Panels (54–21–00).

### Detailed Description

*Refer Figure 54-17*

### Main Side Panels

The side structure of the nacelle center section has upper and lower aluminum alloy outer skins and is attached to the wing lower skin by drag angles. The main landing gear door opening is provided by carbon epoxy sandwich construction, lower side panels attached to lower longerons with a closing aluminum alloy web, form a box section construction.

### Main Landing Gear ’A‘ Frame

The main landing gear “A” frame is machined from solid aluminum alloy and is attached to the wing front spar, upper longeron and the forward nacelle mid−struts.

The bottom of the main landing gear drag strut and side braces are attached to the lower longerons, forward nacelle lower struts with back−up fittings and a shear deck. Support structure is included for the firewall and oil cooler suspension. There are two titanium non−firewalls spaced 1.3 in. (33 mm) apart to create ventilation zone 4, with an inlet for airflow to supply the exhaust forward ejector.

### Firewall and Oil Cooler Support Structure

The oil cooler is installed to a fixed structure at the lower section of the firewall. The mounting structure has two aluminum alloy machined ribs with a shear diaphragm on top. The side ribs also include the forward lower hinge tracks. The back−up structure for the oil cooler is covered by a removable horizontal titanium fire floor, which is sealed against the forward, lower cowl fire door.

### Wing Front and Rear Spar Attachments

The wing rear spar attachments to the nacelle have two fittings machined from aluminum alloy. The attachments supply nacelle to wing, main landing gear yoke and coat hanger interfaces. The attachment fittings are attached to the longerons and mounted off the rear spar. The coat hanger is also attached to the rear spar at its mid−point. The wing front spar attachment is through the A frame integral lugs, with forward and aft, steel fittings, for multi−load path.

### Nacelle Centre Access Panels

The nacelle center access panels give access for maintenance tasks and are machined from aluminum alloy. There are access panels for the wing front and rear spar attachments and the forward exhaust attachments.

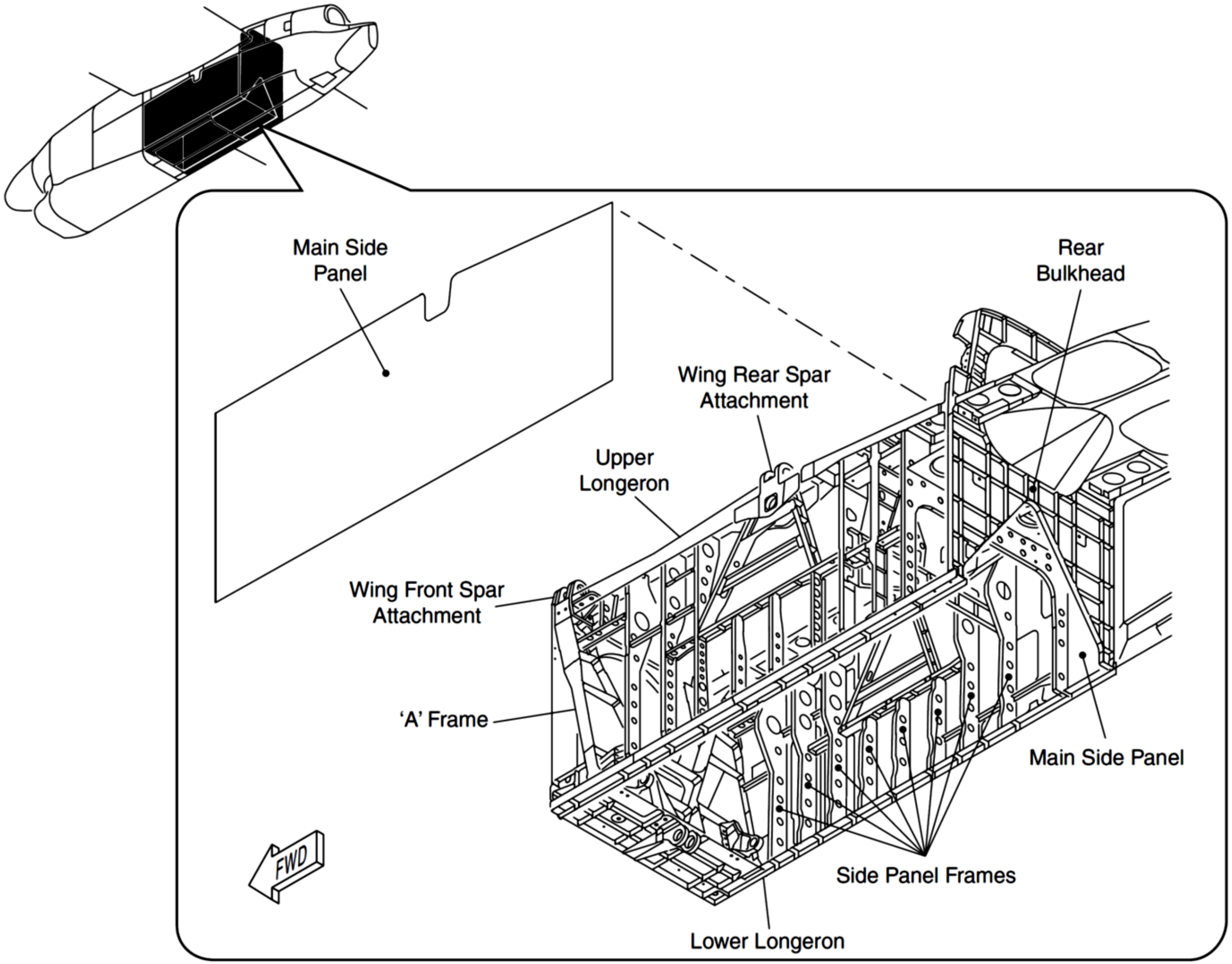



**Figure 54-17 Center Nacelle Frame**

## Center Nacelle Access Panels

### Introduction

The access panels can be removed to do inspection and maintenance tasks.

### General Description

*Refer Figure 54-18*

There are six access panels on the nacelle center structure. The nacelle center structure has the access panels that follow:

- 2 Rear Spar Access Panels
- 2 Front Spar Attachment Access Panels
- 2 Forward Exhaust Attachments Access Panels.

### Detailed Description

*Refer Figure 54-19*

### Rear Spar Access Panel

These access panels are machined from solid aluminum alloy. The access panels have sixteen countersunk holes and are attached to the nacelle center structure. The panels are removed to gain access to the rear spar attachments.

### Front Spar Attachment Access Panel

These access panels are machined from solid aluminum alloy. The access panels have thirteen countersunk holes and are attached to the nacelle center structure with bolts. The panels are removed to gain access to the forward spar attachments.

### Forward Exhaust Attachments Access

The access panels are machined from solid aluminum alloy. The inboard access panels have thirteen holes and are attached to the nacelle center structure with bolts. The panels are removed to gain access to the forward exhaust attachments.

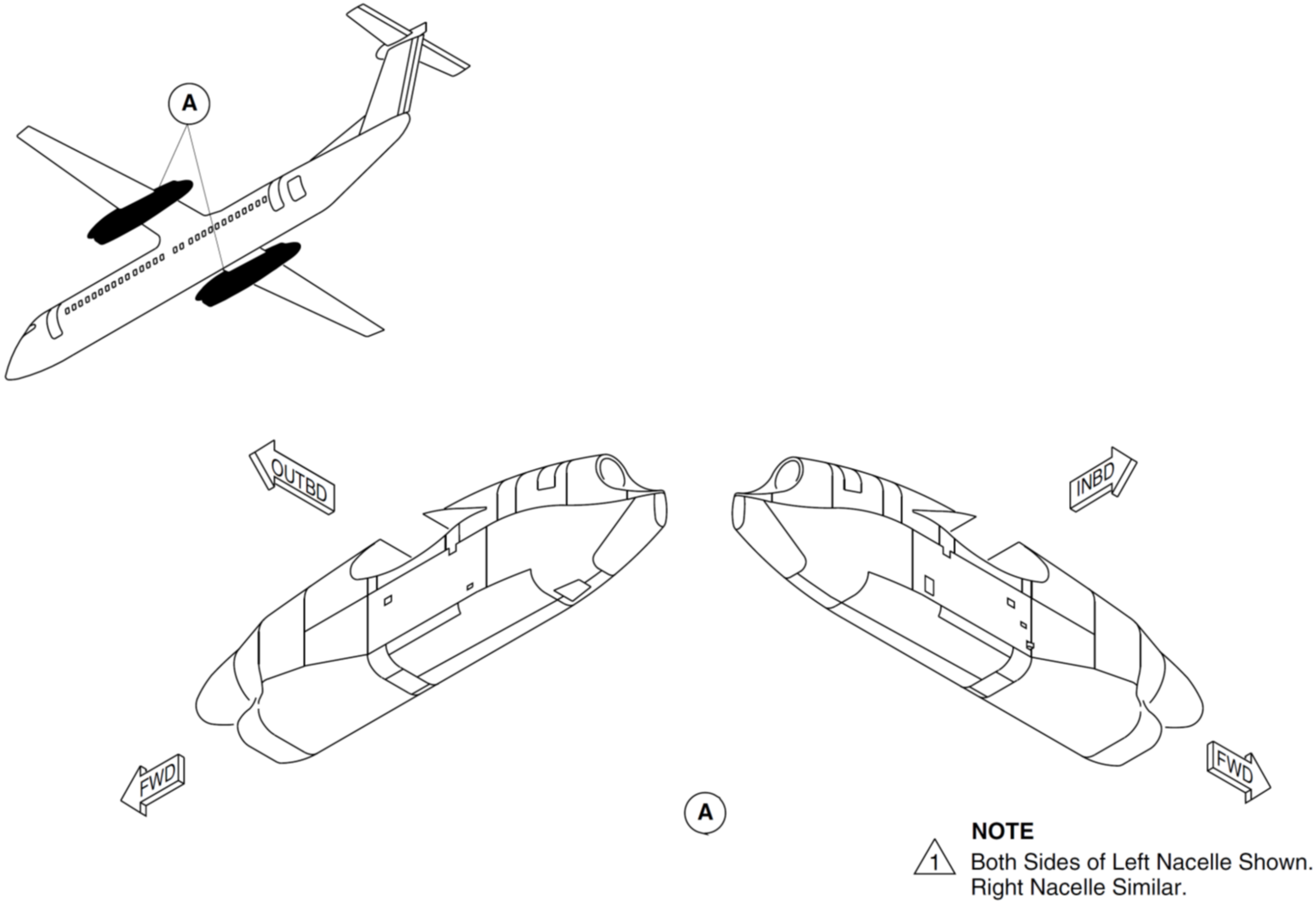



**Figure 54-18 Spar/Forward Exhaust Access Panels - Location**

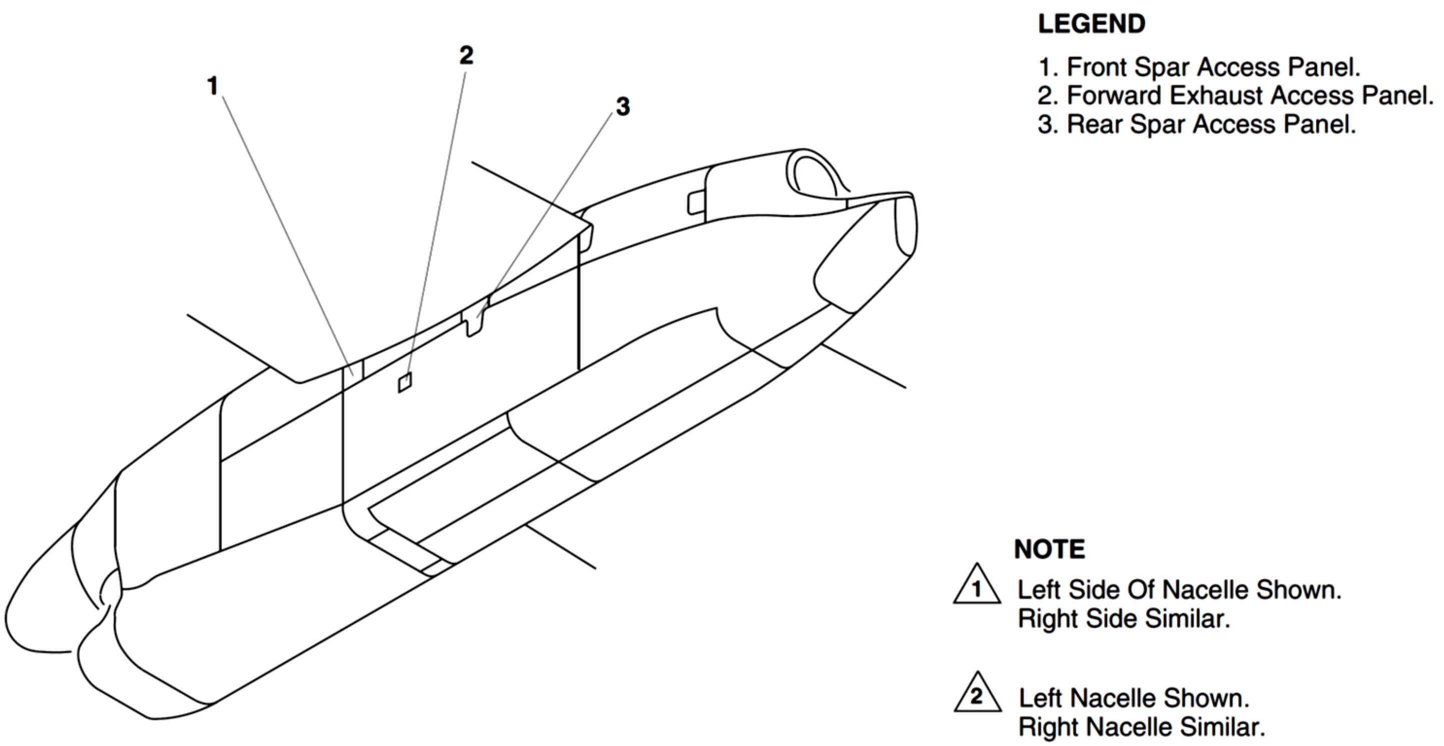



**Figure 54-19 Spar/Forward Exhaust Access Panels**

# Aft Nacelle and Access Panels

## Introduction

The nacelle aft section houses part of the exhaust assembly, and the main landing gear wheels when it is retracted. It includes the structure of the lower fairing and the forward and aft upper fairings.

## General Description

The nacelle aft section has the parts that follow:

- Nacelle Aft Access Panels
- Nacelle Aft Fairings and Surfaces

## Nacelle Aft Access Panels

*Refer Figure 54-20*

The nacelle aft access panels/doors are removed/opened to do maintenance tasks and inspections.

There are three access panels in each nacelle aft section. The two aft exhaust trunnion access panels are located on each side of the nacelles. They gives access to the trunnion bearing.

## Aft Exhaust Trunnions Access Panels

An access panel is installed on each side of the nacelle aft section at the mid upper position. The access panels are machined from solid aluminum alloy. The access panels have seven countersunk holes and are attached to the aft mid upper cowl and the nacelle aft structure with bolts.

## Flaps Access Panel

In each engine nacelle, the aft upper inboard area is installed with a flap access panel.

The access panel is installed on the inboard side of the nacelle aft section at the forward upper position. The access panel is machined from solid aluminum alloy. The access panel has 24 countersunk holes and is attached to the aft mid upper cowl and the nacelle aft structure with bolts.

The flap access panel gives the access to the inspection and maintenance of the flap−track mounting structure.

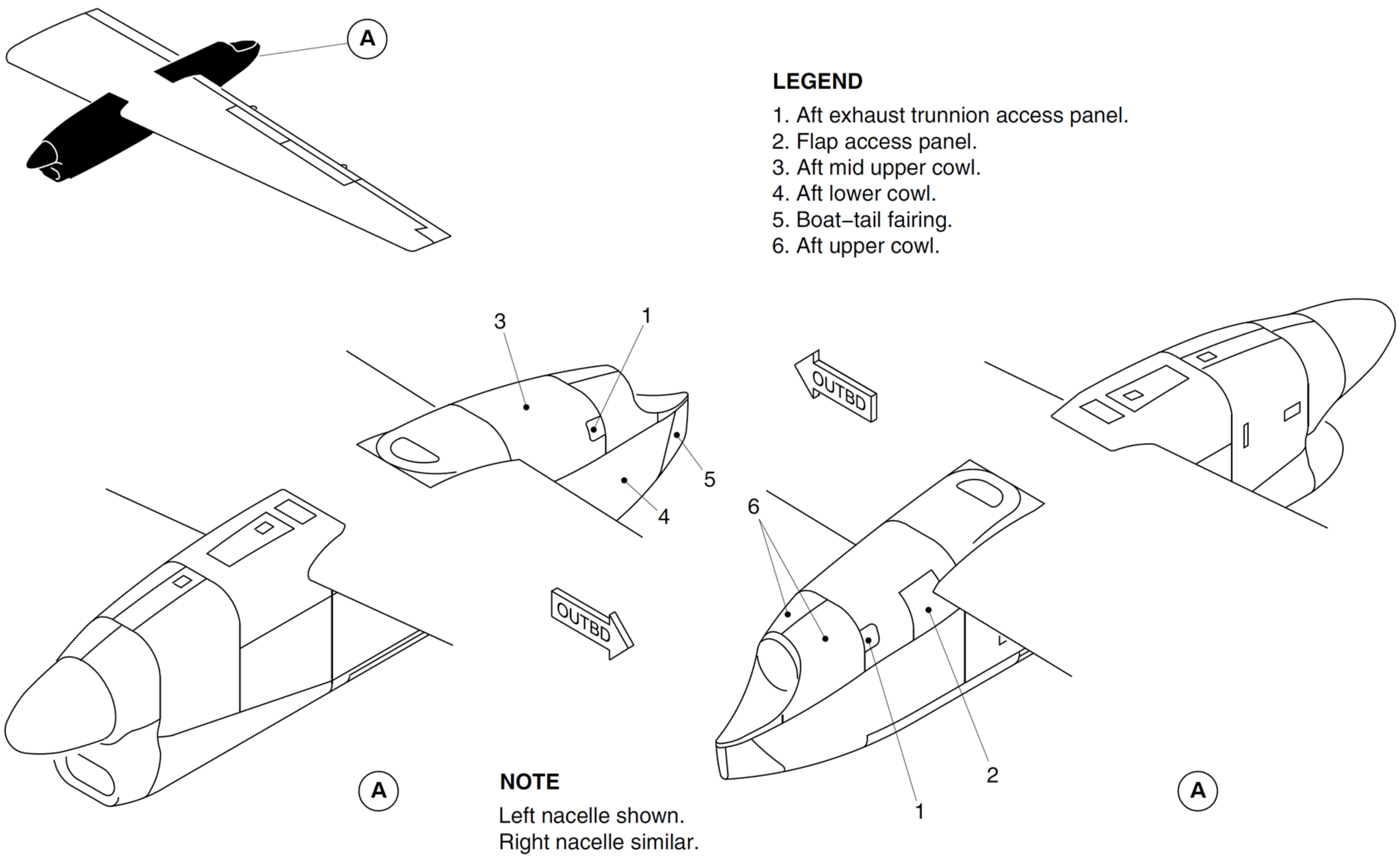



**Figure 54-20 Aft Nacelle Cowl**

**Refuel/Defuel Panel Access Door (Right Nacelle)**

Refer to Figure 54-21

The access door is installed on the bottom of the right aft nacelle. The access door is machined from solid aluminium alloy. The access door opens out on a hinge at its forward edge, and latches in the closed position at two points on its rear edge. There are ten countersunk holes in the aft lower cowl for attachment by bolts to the access door hinge.

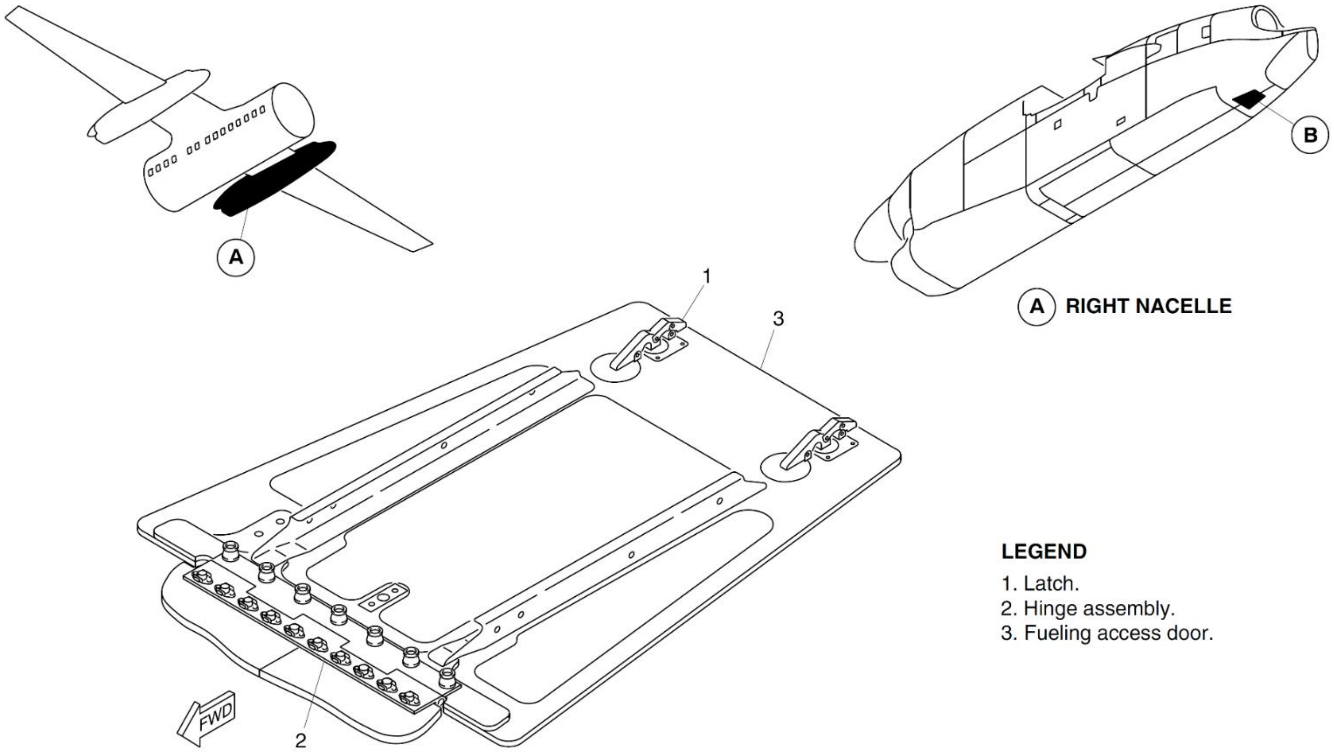



**Figure 54-21 Fueling Access Door – Right Aft Nacelle**

# Aft Nacelle Fairings and Surfaces

## Introduction

The nacelle aft fairings give an aerodynamic surface to the aft part of the engine exhaust and the retracted Main Landing Gear (MLG) wheels. The nacelle aft top fairings can be removed to give access to the engine exhaust.

## General Description

*Refer Figure 54-22*

The nacelle aft section has fairings and surfaces in the areas that follow:

- Top Part of the Nacelle Aft Section
- Bottom Part of the Nacelle Aft Section.

## Detailed Description

*Refer Figure 54-23*

### Top Part of the Nacelle Aft Section

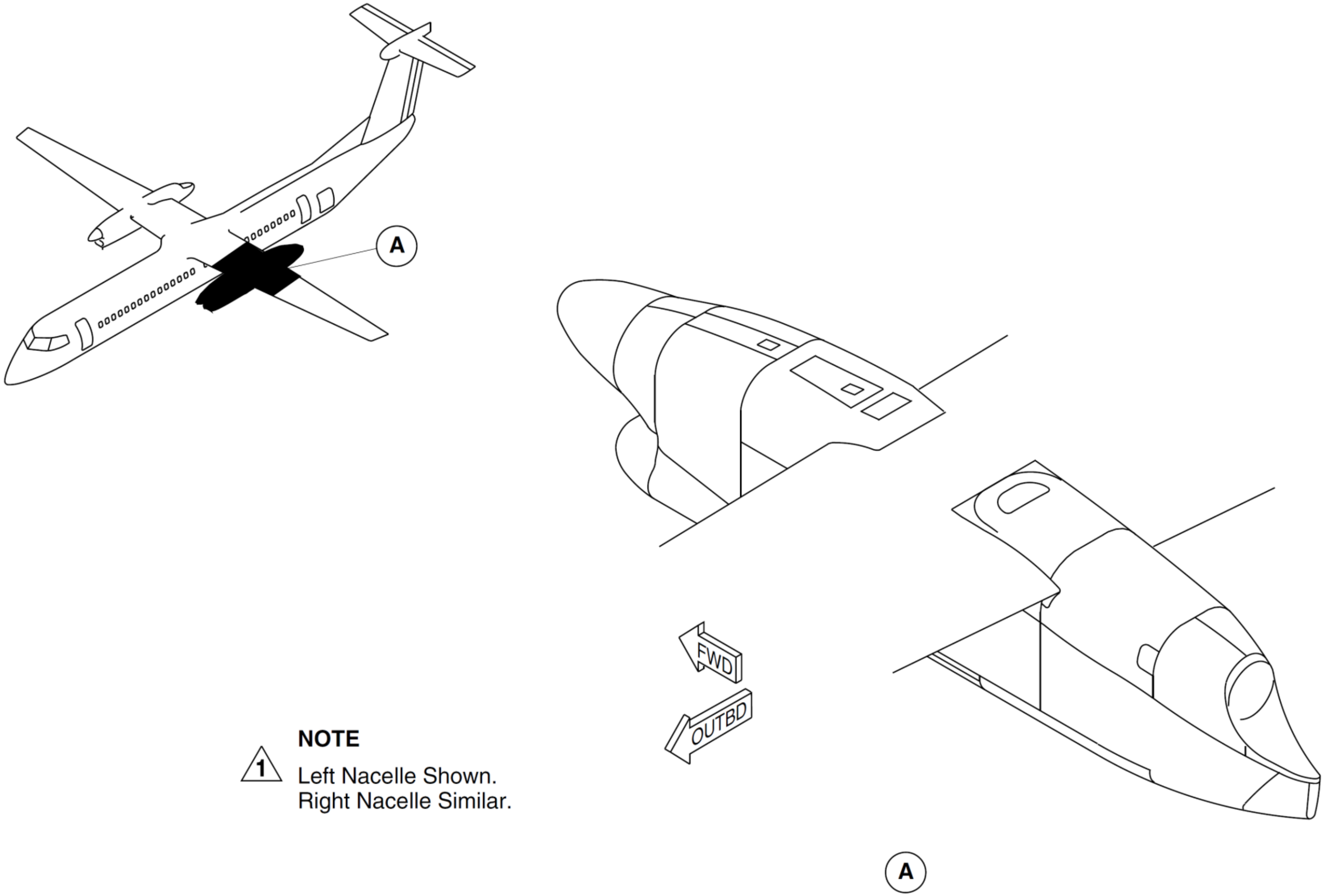
The top fairing is located above the aft exhaust assembly and has two parts:

- The aft mid upper cowl (forward part) is a ’U’ shaped fairing made from carbon fibre/nomex honeycomb. It is attached at its forward end to an alloy frame at the rear of the nacelle center section.
- The aft upper cowl (aft part) has two skins made from fabricated aluminum alloy. This makes the opening at the rear for the aft exhaust outlet. The aft upper cowl gives a clearance around the aft exhaust outlet, which lets a secondary airflow decrease the exhaust temperature in the nacelle.

### Bottom Part of the Aft Nacelle

The bottom part of the nacelle aft section has a large panel (aft lower cowl) on each side with a bulkhead that connects them at the midpoint. They are made from carbon fibre/nomex honeycomb. Expanded copper foil is put in the outer surface of the aft lower cowl. The front end of the aft lower cowl makes the shape of a fairing for the retracted MLG wheel unit. The rear end of the aft lower cowl (right nacelle only) makes the shape of a fairing for a refuel/defuel adapter and control panel.

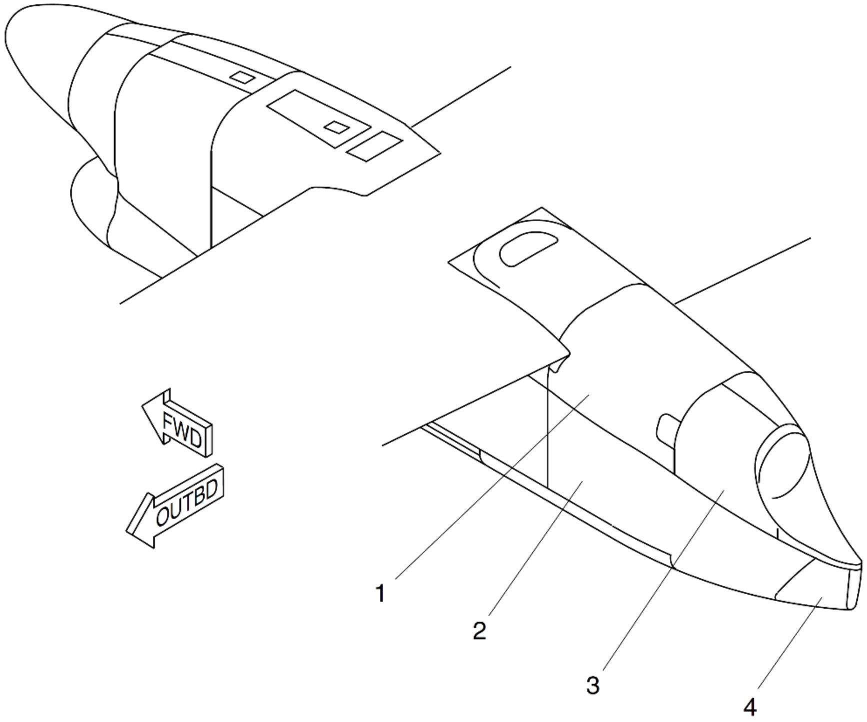
The aft lower cowl is closed at the rear end by the boat−tail fairing. The top of the aft lower cowl is closed by a heatshield. The heatshield is made from fabricated aluminum alloy, part of which is in the shape of the MLG wheels when in the retracted position.




**NOTE**

1 Left Nacelle Shown.
Right Nacelle Similar.

**Figure 54-22 Aft Nacelle Fairings and Surfaces - Location**




**LEGEND**

1. Aft Mid Upper Cowl.
2. Aft Lower Cowl.
3. Aft Upper Cowl.
4. Boat–Tail Fairing.

**NOTE**

1 Left Side of Nacelle Shown.
Right Side Similar.

**Figure 54-23 Aft Nacelle Fairings and Surfaces**